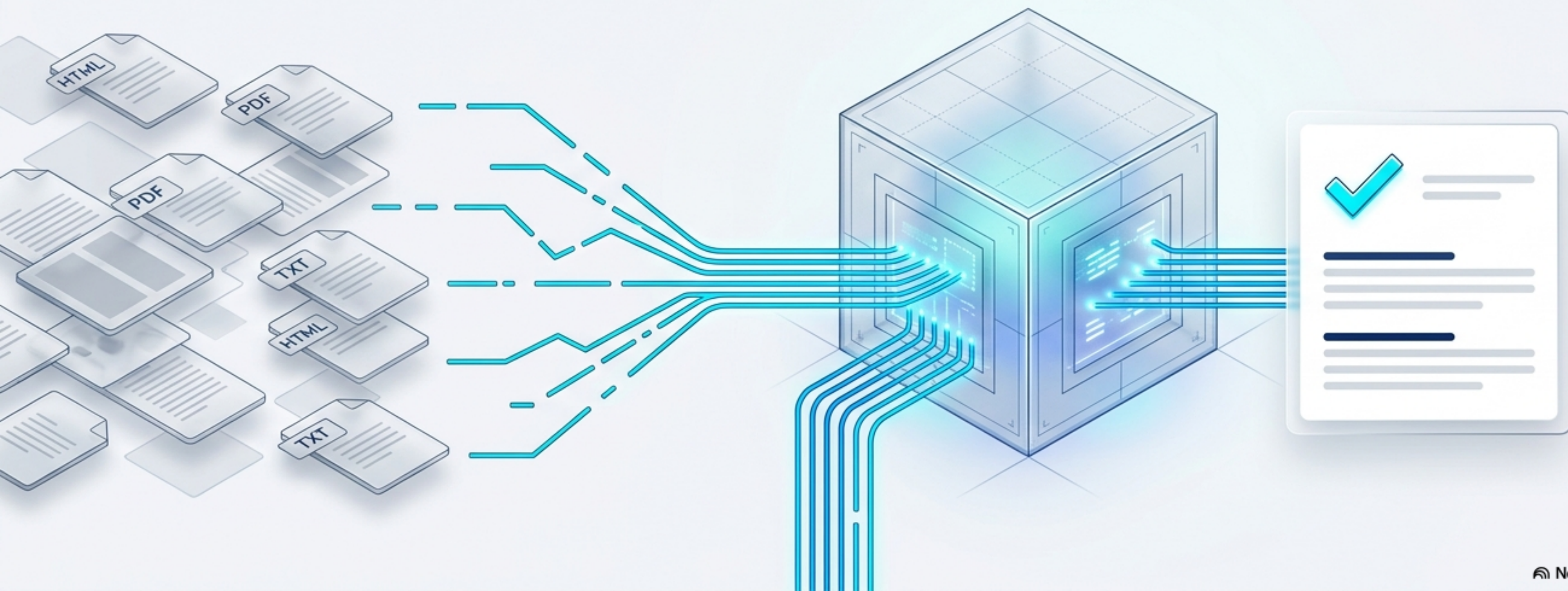
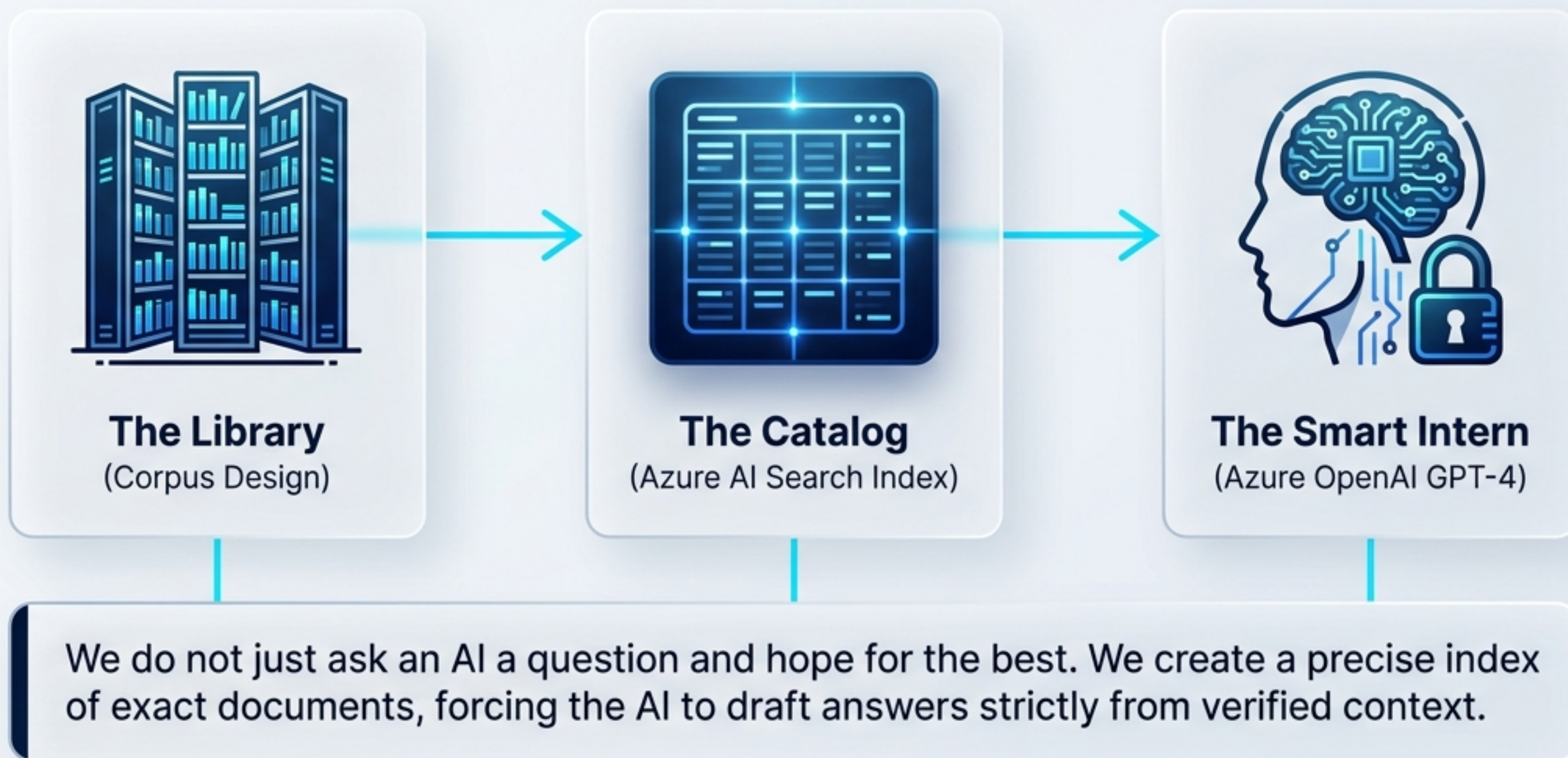


The AskTelstra RAG Engine.

Turning unstructured enterprise knowledge into instant, accurate, and fully grounded answers.



We build a highly structured catalog to ground the AI.

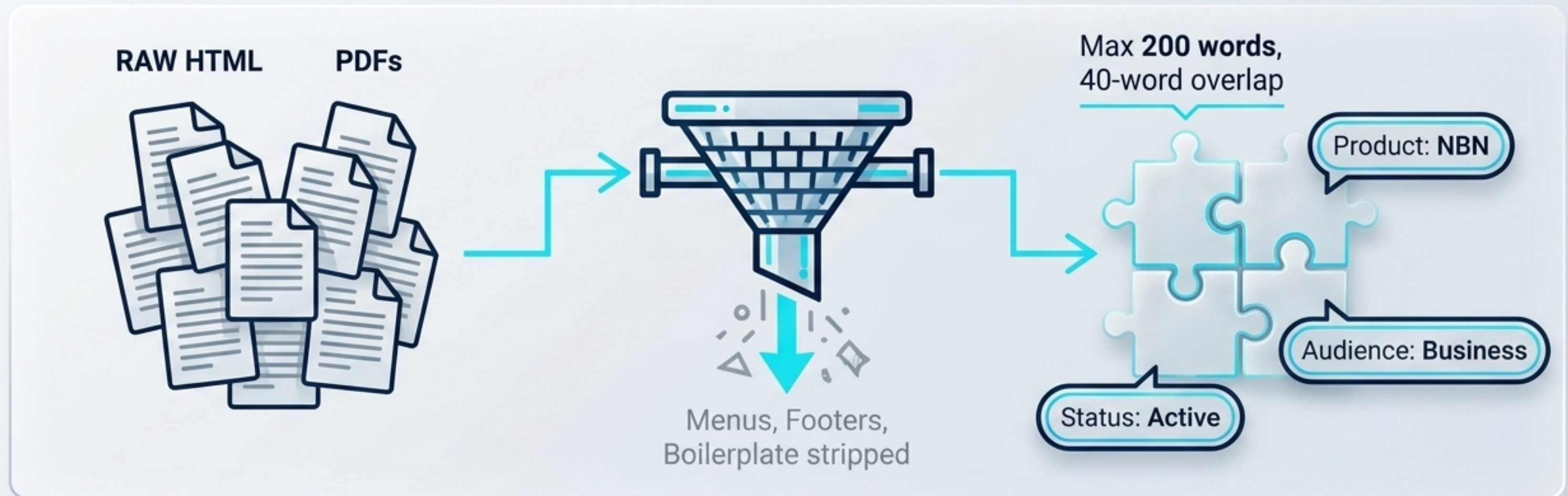


Five sequential stages guarantee grounded, accurate AI responses



Aggressive cleaning and tagging transform raw data into semantic blocks.

Before the AI processes anything, documents are stripped of noise, divided into coherent mini-cards, and injected with hard metadata to ensure precise filtering.



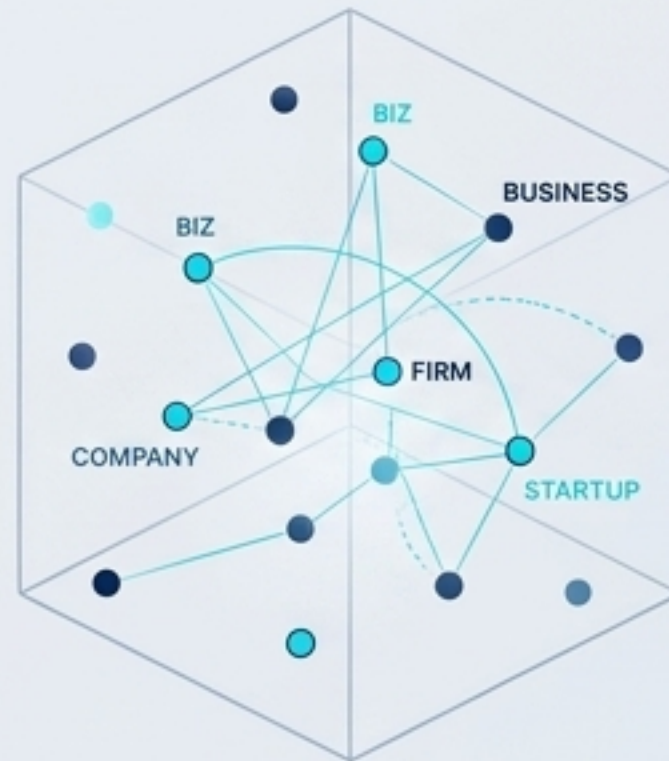
Vector search combined with hard metadata rules captures true user intent

Keyword Search (Legacy)



Relies on exact phrasing.
Fails on synonyms.

Semantic Vector Search (FAISS)



Converts text to mathematical vectors to understand intent (e.g., 'biz' = 'business').

Filtered Vector Schema (Target)



Semantic search constrained by hard metadata rules (e.g., excluding 'deprecated' plans).

The generation layer acts as an ultimate defense against hallucinations.

The Large Language Model is confined strictly to the retrieved context. If the answer is not in the source data, the system is programmed to admit it does not know.



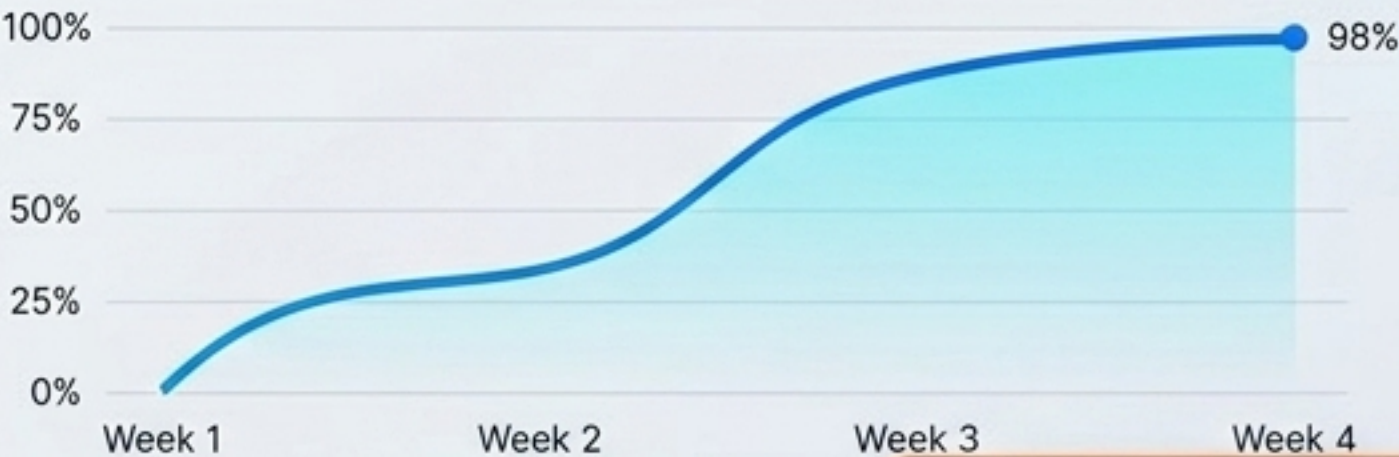
Continuous testing against a 'Gold Set' guarantees search quality.

We aggressively test the pipeline against pre-validated queries to monitor retrieval precision, track answer correctness, and enforce document freshness.

Precision@3



Correctness Rate over time



Freshness Rule Triggered:
Chunk > X days old

Gold Set Evaluation

Query	Expected Answer	Tag	Correct Answer Flag
Query: How to reset password	Navigate to settings...	User Account	✓
Query: Explain API rate limits	See developer docs...	Developer API	✓
Query: Contact support team	Email support@...	General Support	✓
Query: How to report a bug	Navigate to settings...	General Support	✓

A local prototype systematically scales into robust enterprise architecture.

The core operational pattern remains identical as we graduate from local Python scripts to fully orchestrated Microsoft Azure and Fabric enterprise environments.

Prototype (Laptop)

- Python
- Local Folders
- FAISS Vector Store
- OpenAI

Staging (Semi-Realistic)

- CSV/Logs
- Azure Blob Storage
- Azure AI Search free tier

Enterprise (Telstra-Shaped)

- Fabric Pipelines
- Synapse Notebooks
- Azure AI Search schemas
- Power BI Dashboards

The fully integrated, Azure-native ecosystem is ready for deployment.

A secure, highly scalable cloud architecture designed to map perfectly to enterprise APIs, data controls, and reporting standards.

